



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES PAPER 1	DATE	21 JULY 2023
GRADE	11	MARKS	150
EXAMINER NAME	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
CLASS		DURATION	2 ½ HOURS

INSTRUCTIONS:

1. This paper consists of 24 pages- 21 pages of questions and 3 pages of additional writing space. Please check that your question paper is complete.
 2. Answer all questions on the question paper.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. Any calculations need to be rounded off to TWO decimal places unless stated otherwise.
 7. It is in your own interest to write legibly and to present your work neatly.
-

QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	60	
2	30	
3	30	
4	30	
TOTAL: 150		

Question 1:

- 1.1. Select the name/ or term in Column II which best matches the description in Column I. Write the letter of the matching name/term in the appropriate space provided between the brackets. Each letter may only be used once. [10]

<u>Column I</u>		<u>Column II</u>	
[]	Acellular	A	bacillus
[]	Spores	B	protists
[]	Spherical shape	C	asexual
[]	Consumers	D	virus
[]	Binary fission	E	plasmid
[]	Circular ring of DNA	F	coccus
[]	Eukaryotic	G	plankton
[]	Plant-like protists	H	fungi
[]	Floating organisms	I	algae
[]	Saprotrophic	J	autotrophic
		K	heterotrophic
		L	decomposers

LIFE SCIENCES

1.2. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question in the space provided in the table below:

1.2.1.	1.2.2.	1.2.3.	1.2.4.	1.2.5.	1.2.6.	1.2.7.	1.2.8.	1.2.9.

1.2.1. Antibiotics are used to treat some diseases.
Which diseases could they be effective against?

	Bacterial infection	HIV	Yeast infection
A	✓	✓	✓
B	✓	x	✓
C	✓	x	x
D	x	✓	x

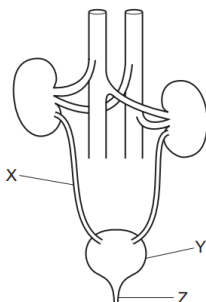
(1)

1.2.2. Viruses consist of:

- A. RNA, DNA and a protein coat
- B. Proteins, cell membrane and RNA
- C. RNA or DNA and a cell membrane
- D. RNA or DNA and a protein coat

(1)

1.2.3. The diagram below shows the human excretory system and related blood vessels.



What are structures X, Y and Z?

	X	Y	Z
A	ureter	bladder	urethra
B	ureter	kidney	urethra
C	urethra	bladder	ureter
D	urethra	kidney	ureter

(1)

1.2.4. Which row describes the urine produced by a person on a hot day?

	Concentration of urine	Volume of urine
A	concentrated	large
B	concentrated	small
C	dilute	large
D	dilute	small

(1)

1.2.5. In healthy people, which substance is completely reabsorbed into the blood from the kidney tubules?

- A. Glucose
- B. Salts
- C. Urea
- D. Water

(1)

1.2.6. The axial skeleton is made up of the following regions:

- A. Skull, vertebral column and pelvis
- B. Skull, vertebral column, ribs and sternum
- C. Skull, pectoral girdle, ribs and sternum
- D. Skull, pelvic girdle, ribs and sternum

(1)

1.2.7. Which of the following statements is NOT true about tendons?

- A. Connects muscle to bone.
- B. Comprised of collagen and elastic fibres.
- C. Stabilizes joints.
- D. Bring about movement.

(1)

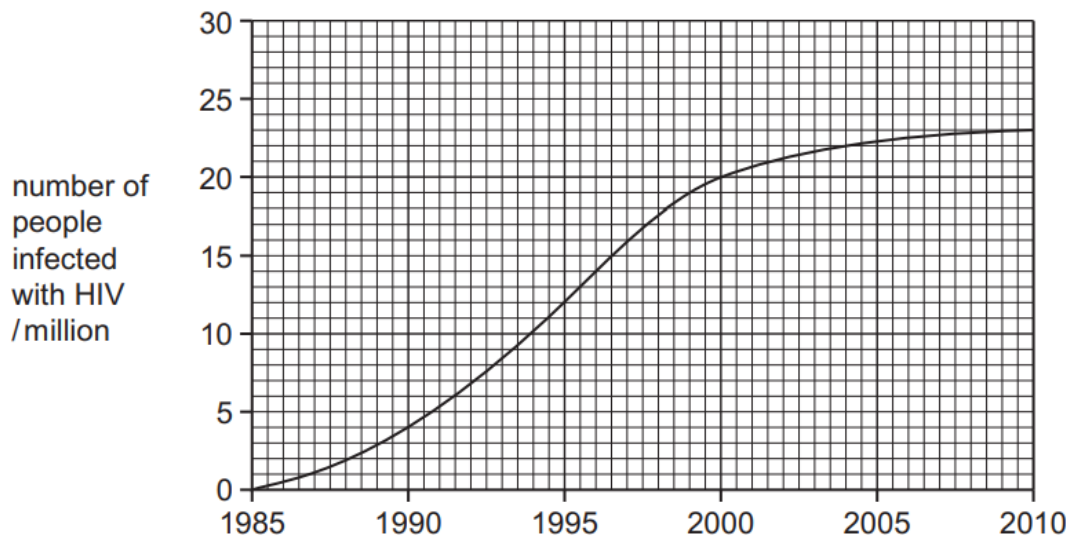
1.2.8. What type of bone would a rib be classified as?

- A. Long bone
- B. Short bone
- C. Flat bone
- D. Irregular bone

(1)

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- 1.2.9. The graph shows the number of people infected with HIV, in one part of the world, from 1985 to 2010. (2)



Using data from the graph, which statement is correct?

- A. Between 1995 and 2000 the number of people infected with HIV increased by 67%.
- B. Between 1995 and 2000 the number of people infected with HIV increased by 20%.
- C. Between 1990 and 1995 the number of people infected with HIV doubled.
- D. Between 1995 and 2000 the number of people infected with HIV doubled.

[10]

- 1.3. HIV is a pathogen that can cause AIDS. The table below shows the adult population size of a country and the estimated number of people infected with HIV/AIDS in that country in 2016. Data for six countries are shown.

Country	Adult population size	Estimated number of adults infected with HIV/AIDS
A	808 824	220 000
B	248 490	8 200
C	221 000	221
D	3 250 000	130 000
E	5 111 111	46 000
F	1 333 333	48 000

- 1.3.1. What is a pathogen? (1)

- 1.3.2. Which country had the greatest number of adults infected with HIV/AIDS? (1)

1.3.3. Calculate the percentage of adults from Country B that were infected with HIV/AIDS. (3)

1.3.4. How could the government of countries greatly affected by HIV/AIDS reduce the number of people infected in the future? (2)

1.3.5. In the space below draw a bar graph to show the estimated number of adults infected with HIV/AIDS. (7)

LIFE SCIENCES

1.4.

1.4.1. The box on the left shows the beginning of a sentence. The boxes on the right show some sentence endings.

Draw TWO straight lines from “Excretion” to the boxes on the right to make two correct sentences.

(2)

Excretion	is the movement of digested food molecules into cells.
	is the passing out of undigested food from an organism.
	is the removal of excess substances from an organism.
	is the removal of toxic materials from an organism.
	is the taking in of materials for energy and growth.

1.4.2. Name TWO excretory organs, other than the kidneys, found in the body and what substance they excrete.

(4)

1.4.3. The kidneys filter blood and produce a liquid known as urine. Name three substances found in urine.

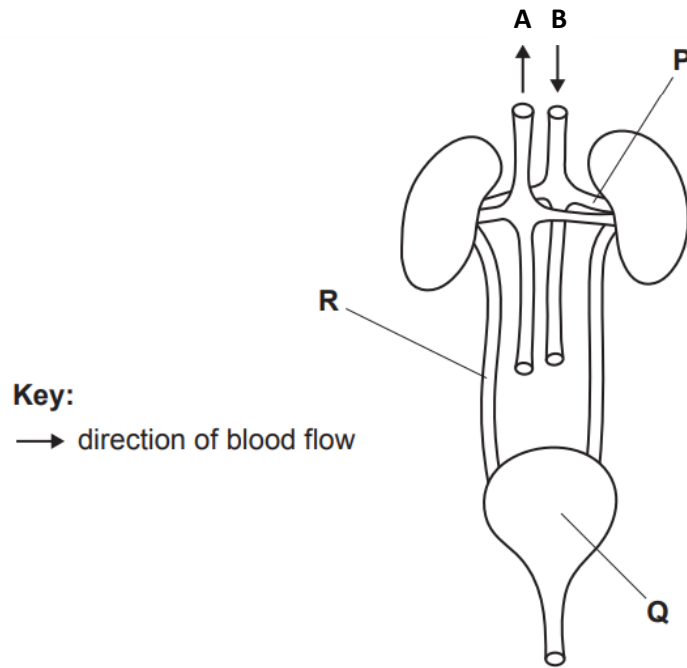
(3)

1- _____

2- _____

3- _____

1.4.4. Below is a diagram of part of the human excretory system and associated blood vessels.



A. Name structures P, Q and R.

(3)

P- _____

Q- _____

R- _____

B. Tabulate two differences between the blood found in vessel A and vessel B.

(5)

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- 1.4.5. The volume and concentration of urine produced is affected by changes in water consumption, temperature and exercise levels.

The table below shows three different conditions.

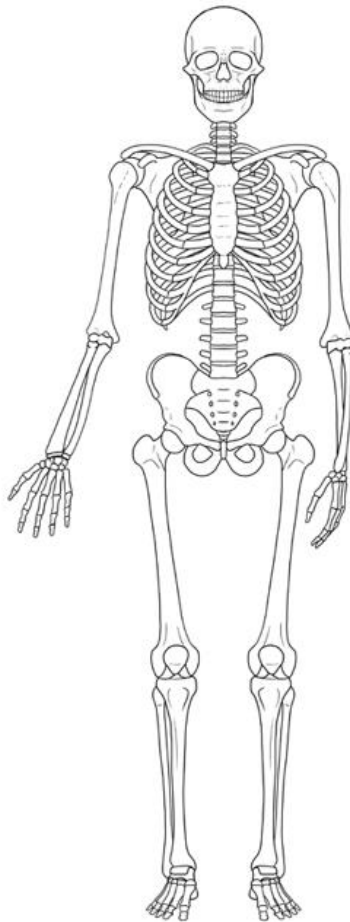
Complete the table by writing **increases or decreases** in the spaces to show the effect of each condition on the volume and concentration of urine produced. (3)

Condition	Volume of urine	Concentration of urine
Increase in water consumption		
Increase in temperature		
Increase in exercise level		

[20]

- 1.5. In the word box below are the names of six different bones. Label the bones on the diagram of skeleton.

Cranium	Clavicle	Radius	Femur	Sacrum	Tarsals
----------------	-----------------	---------------	--------------	---------------	----------------

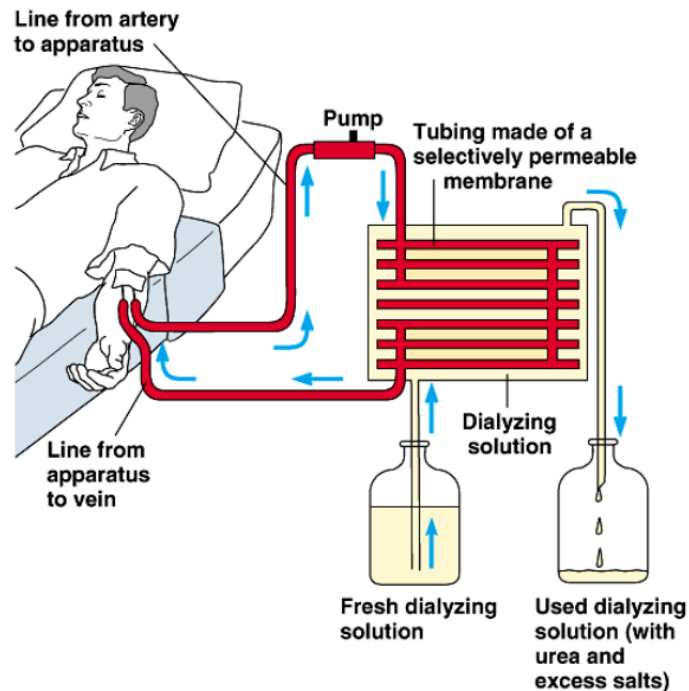


[6]

Total Question 1: 60 marks

Question 2

- 2.1. If a person's kidneys happen to get damaged or do not function correctly a person can be placed on a dialysis machine to help the functioning of the kidneys.



- 2.1.1. Define the term excretion.

(2)

- 2.1.2. In which part of the nephron does a similar process take place as illustrated by the kidney machine?

(1)

- 2.1.3. Compare the composition of the dialysis fluid and the patient's blood.

(4)

2.1.4. Give a reason for your answer to QUESTION 2.1.3.

(2)

2.1.5. Explain why dialysis fluid is continually replaced.

(2)

2.1.6. Explain the purpose of having so many tubes carrying blood through the dialyser.

(2)

2.1.7. Provide two disadvantages of dialysis.

(2)

2.1.8. Dialysis is a short-term solution to kidney failure. What is the long-term solution?

(1)

[16]

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2.2. Figure A is a diagram of a filtration unit of the kidney. The arrows show the direction of blood flow. Figure B is a drawing of a vertical section through a cell from the lining of region 2 of diagram A.

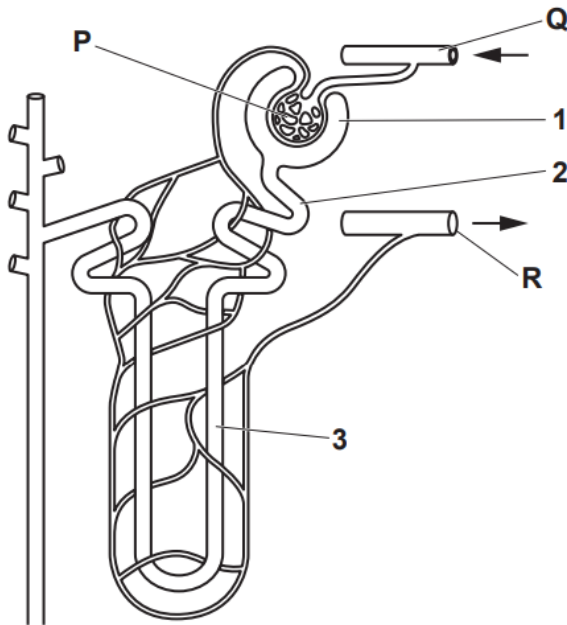


Figure A

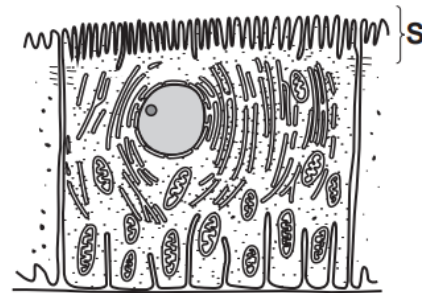


Figure B

2.2.1. Provide the name of the kidney unit seen in Figure A. (1)

2.2.2. Structure R is a branch of which main vessel? (1)

2.2.3. Provide the name of structure P. (1)

2.2.4. In which region of the kidney would you find structure 1 and structure 3? (2)

1- _____

3- _____

2.2.5. What is the name of region 2 where the cells in Figure B are found? (1)

2.2.6. Explain why the cells in Figure B contain many mitochondria.

(2)

2.2.7. Below are the concentrations of some substances in blood plasma and in the regions labelled 1 and 3 on Figure A.

Substance	Concentration/ mg per cm ³		
	Blood plasma	Region 1	Region 3
Protein	8000	0	0
Glucose	100	100	0
Salts	320	320	300
Urea	30	30	2000

Explain the difference in concentration of the following:

a. Protein in blood plasma and region 1.

(2)

b. Glucose in region 1 and region 3.

(2)

c. Urea between region 1 and region 3.

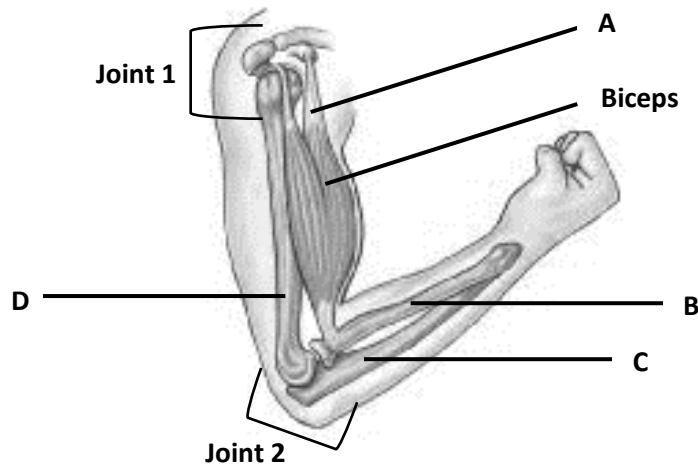
(2)

[14]

Total Question 2: 30 marks

Question 3

3.1. The diagram below shows the bones and muscles in the human arm.



3.1.1. Identify bones B, C and D. (3)

B- _____

C- _____

D- _____

3.1.2. Name structure A and provide it's function. (2)

3.1.3. Explain the treatment necessary if structure A were to rupture. (2)

3.1.4. Joint 1 and 2 are both types of synovial joints. In the space **on the next page** tabulate the difference between them by comparing the following: (7)

- Name of synovial joint
- Type of movement allowed
- Where it is found in the body (other than in the diagram above)

3.1.5. Various type of diseases are associated with the skeletal system. Identify the disease characterized by each of the following:

A. bowing of the legs: _____

B. weak and brittle bones: _____

C. inflammation and pain at the joints: _____

(3)

[17]

3.2. Use the following information on Rheumatoid arthritis and your own knowledge to answer the questions that follow.

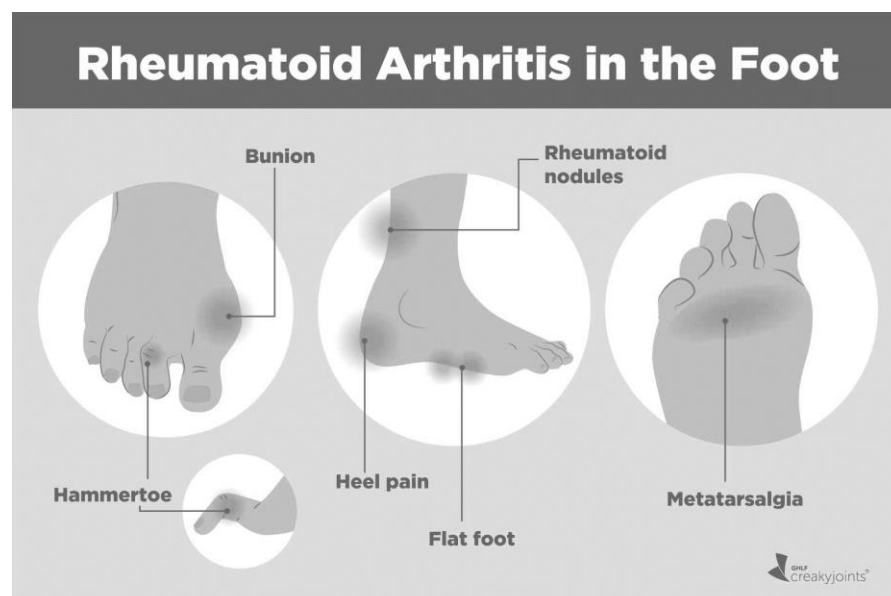
Rheumatoid arthritis (RA) is a chronic, inflammatory disorder in which your immune system mistakenly attacks the lining of membrane that surrounds your joints. It tends to target smaller joints first, such as those in your hands- and often, your feet. In fact, more than 90 percent of people with RA will develop pain, stiffness, swelling or other symptoms in the foot and ankle over the course of the disease, according to the American Academy of Orthopaedic Surgeons.

How Rheumatoid Arthritis affects the feet

Joints are covered with a lining called synovium. Its job is to lubricate the joint so it moves more easily. RA causes an overactivity in the lining. The synovium becomes inflamed, thickens, and produces an excess of joint fluid. That extra fluid- along with the inflammatory chemicals released by the immune system- causes swelling, damages cartilage, and softens the bone within the joint. A systemic disease, RA also affects ligament and surrounding soft tissues, says Brett Sachs, DPM, foot and ankle surgeon in Colorado. “When that happens, the joints start to weaken and that’s when the deformities can occur,” explains Dr. Sachs, who is also a fellow member of the American College of Foot and Ankle Surgeons. Rheumatoid arthritis tends to affect the joints in your feet more quickly in part because they are smaller, says Dr. Sachs. Research shows in about 20 percent of RA patients, foot and ankle symptoms are the first signs of the disease.

What does Rheumatoid Arthritis in the feet feel like?

When your feet are affected by RA, you may experience swelling, redness, and a feeling of warmth around the affected joints. Pain is very common as well. In one study comparing foot problems in RA patients versus people without arthritis, 98 percent of RA patients had foot pain and 96 percent reported some difficulty in function, compared to 76 percent and 66 percent, respectively, in the healthy group. Compared with osteoarthritis, which typically affects one specific joint, RA usually occurs in the same joints on both sides of your body- so it commonly affects both feet at once. Symptoms may flare, then go into periods of remission. Over time, RA can cause joints to deform and shift out of place. Specific foot problems caused by RA include:



3.2.1. What is rheumatoid arthritis? (2)

3.2.2. According to Dr. Sachs, what causes the deformities in the feet of a rheumatoid arthritis sufferer? (2)

3.2.3. What is the “excess joint fluid” referred to in paragraph 2 better known as? (1)

3.2.4. Explain the findings (paragraph 3) in the study that compared foot problems in RA patients and patients without arthritis. (2)

3.2.5. Name TWO fixed variables that the investigation would have had to adhere to. (2)

3.2.6. One of the foot problems caused by RA is a bunion.
Describe what a bunion is. (2)

3.2.7. What is osteoarthritis? (2)

[13]

Total Question 3: 30 marks

Question 4

- 4.1. Garlic is known to have the ability to fight bacteria and viruses. It is known to be effective against a wide range of bacteria and can combat the common cold. The antimicrobial substance in garlic is called *allicin*. To maintain the antibacterial properties of garlic, it must be consumed or applied as raw garlic.

Scientists wanted to investigate the effectiveness of garlic in killing bacteria. They conducted the experiment as follows:

- They used three petri dishes prepared with blood agar and stored these in a refrigerator.
- Before the start of the experiment, they removed the petri dishes from the refrigerator to allow them to reach room temperature.
- They prepared three test specimens and labelled them as described below:
 - The three test tubes were labelled **A**, **B** and **C**.
 - The contents of the test tubes were measured and mixed as shown in the table below:

Test tube	Contents of the test tubes		
	100 mL milk	5 mL <i>E.coli</i> bacterium	Garlic extract
A	✓	x	x
B	✓	✓	x
C	✓	✓	✓

- They petri dishes were labelled **A**, **B** and **C**.
- They removed the lid in petri dish A and used the syringe to extract 10 mL of the sample from test tube **A** and placed it in the center of petri dish A.
- In the same way, using a new syringe a 10 mL sample was extracted from test tube **B** and placed in petri dish **B** and the procedure was repeated for petri dish **C**.
- The petri dish lids were replaced, and the petri dishes were stored in a cool and shaded place.
 - The diameter of the *E.coli* colony was measured every day for 5 days and recorded in the table below:

Petri dish	Diameter of bacteria colony (mm)				
	Day 1	Day 2	Day 3	Day 4	Day 5
A	0	1,7	3,0	4,6	7,1
B	0	4,2	8,4	15,1	36,5
C	0	0	0	0	0

- 4.1.1. Which substance will be destroyed if you were to cook the garlic and then to consume it? (1)

4.1.2. State: (4)

a. The independent variable: _____

b. The dependent variable: _____

c. Two fixed variables: _____

4.1.3. Why were the petri dishes kept in the fridge before the start of the investigation? (2)

4.1.4. Explain the results in the following petri dishes:

Petri dish A: (2)

Petri dish C: (2)

4.1.5. What is the function of the agar when culturing/growing bacteria? (1)

4.1.6. Can garlic be considered an antibiotic? Give a reason for your answer. (2)

4.1.7. In the space **on the next page** draw a labelled diagram of a bacterium. (5)



[19]

- 4.2. A mycorrhiza is a mutualistic relationship between fungal hyphae and the roots of true plants. The hyphae increase the absorptive surface of the plant's roots by aiding in the absorption of water, phosphorus and other mineral ions from the soil to the roots of plants. The plant is photosynthetic and provides the fungus with carbohydrates.

Scientists conducted an experiment to determine the effect of mycorrhizal associations on plant growth. The experiment was conducted as follow:

- Two groups of plants were grown.
- One group was planted in soil that had been sterilised (free from living microorganisms).
- The other group was planted in the same type of soil but the soil had not been sterilised.
- All other factors remained the same between the two groups.
- The plants were allowed to grow for 8 week.
- Each week, the height (in centimeters) of each plant was measured.

The table below shows the growth of the plants over the 8-week period.

Week	Height of plants grown in sterilised soil (cm)	Height of plants grown in non-sterilised soil (cm)
1	0,8	2,0
2	1,5	5,5
3	2,0	8,7
4	2,3	10,0
5	2,4	12,0
6	3,8	16,2
7	5,0	19,1
8	6,0	25,0

4.2.1. What is meant by a mutualistic relationship? (1)

4.2.2. Describe the mutualistic relationship between the fungus and the plants. (2)

4.2.3. Describe ONE way in which the scientists ensured that the results of this investigation are valid. (1)

4.2.4. Explain why the plants grown in sterilised soil grew much slower than plants grown in non-sterilised soil. (3)

4.2.5. What conclusion can be drawn from the results of this experiment? (2)

4.2.6. Name AND explain one other type of symbiotic relationship. (2)

[11]

Total Question 4: 30 marks

TOTAL: 150 MARKS

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